Investment Decision-Making under Market Uncertainty using Statistical Modelling and Bayesian Decision Theory

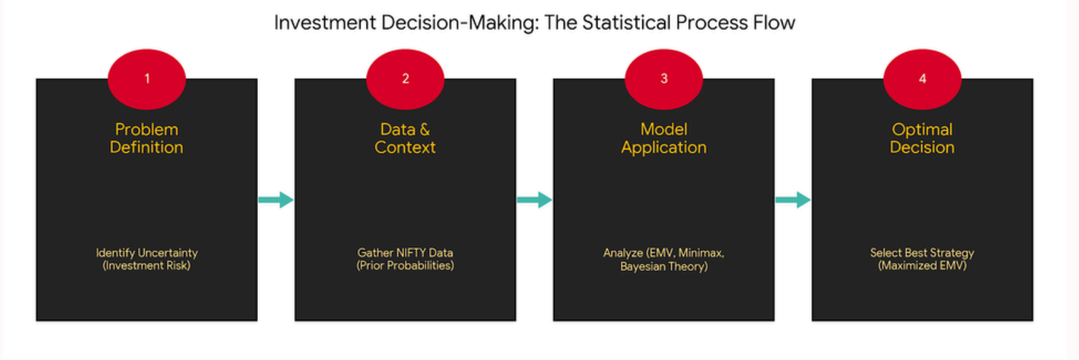
A Case Study on NIFTY 50 (2000–2025)

"This study explores how investors can make rational, data-backed decisions using real NIFTY-50 data from 2000–2025."

Presented by: L074 L056 L043

INTRODUCTION

- Every business and investor faces uncertainty outcomes are not always known.
- Decision-making means choosing the best possible action among alternatives using data and logic.
- Statistical models help reduce uncertainty by using probabilities and expected outcomes.
- In financial markets, investors use decision-making tools to select the most profitable strategy with minimum risk.



What is Decision Making?

Decision-making is the process of selecting the best alternative after analyzing available information and outcomes.

Key Features:

- Involves identifying a problem and choosing between alternatives.
- Aims to reduce risk and uncertainty.
- Requires logical, data-driven thinking.

Linking Decision Theory to Investment

- Financial markets are uncertain; prices move due to many unpredictable factors.
- Statistical Decision Theory allows investors to make rational decisions using:
  - Probability of each market condition
  - Expected payoffs for each action
  - Learning from new data (Bayesian theory)

Goal: Choose the investment strategy that gives best average return under uncertainty.

Problem Statement

"How can an investor select the most profitable and consistent investment strategy under uncertain market conditions using decision-making models such as Expected Monetary Value (EMV), Utility, and Bayesian Updating?"

<u>Objectives:</u>
To identify the best investment strategy among alternatives.
To evaluate decisions under different market states (Bull, Bear, Neutral).
To apply Bayesian theory to update beliefs based on new information.

Date	Open	High	Low	Close
date of trade	stock price on that day when market opens	highest stock price on that day	lowest stock price on that day	stock price on that day when market closed
2000-01-03 2025-05-26	853 26.2k	877 26.3k	850 26.2k	854 26.2k
2000-01-04	1594.4	1641.95	1594.4	1638.7
2000-01-05	1634.55	1635.5	1555.05	1595.8
2000-01-06	1595.8	1639.0	1595.8	1617.6
2000-01-07	1616.6	1628.25	1597.2	1613.3
2000-01-10	1615.65	1662.1	1614.95	1632.9
2000-01-11	1633.25	1639.9	1548.25	1572.5
2000-01-12	1572.3	1631.55	1571.7	1624.8
2000-01-13	1627.85	1671.15	1613.65	1621.4
2000-01-14	1622.15	1627.4	1591.4	1622.7
2000-01-17	1623.5	1668.45	1604.65	1611.6
2000-01-18	1611.65	1615.15	1587.85	1606.7
2000-01-19	1610.05	1644.45	1608.85	1634.8
2000-01-20	1634.65	1644.4	1596.65	1601.1

Nifty 50 (2000–2025) CSV (Kaggle)

Dataset Used:

NIFTY 50 Index Historical Data (2000–2025)
Source: Kaggle (Backup: NSE / Yahoo Finance)
Data Type: Daily closing prices of the NIFTY 50 Index
Period Covered: 25 years (January 2000 – October 2025)
Total Records: ~6,000 daily observations

Derived Metrics:

- Monthly Start Price
- Monthly End Price
- Monthly Return (%)

Formula Used:

$$\text{Monthly Return (\%)} = \frac{P_{end} - P_{start}}{P_{start}} \times 100$$

where Pstart and Pend are the first and last closing prices of each month.

Dataset

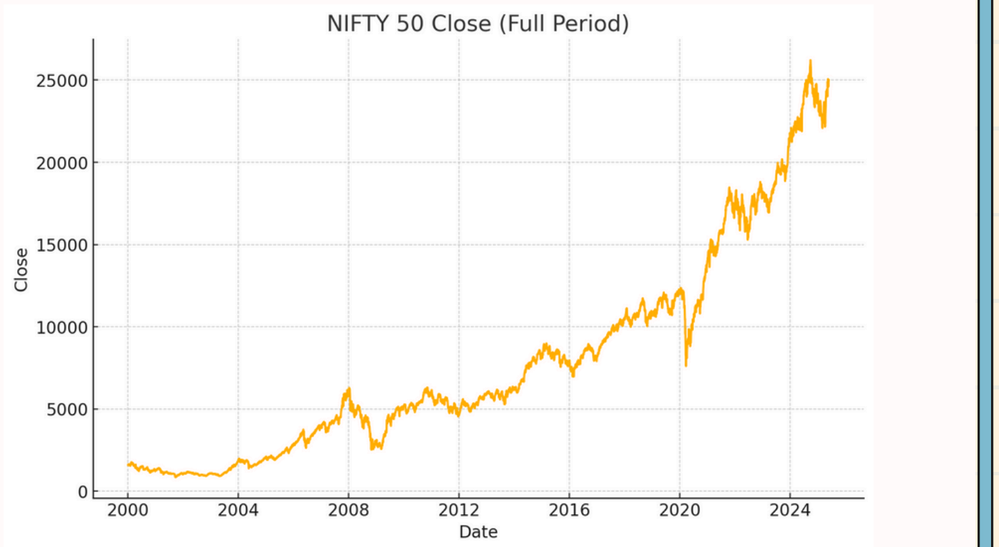
Market State Classification

Market State	Count (Months)	Probability
Bull	150	0.50
Bear	110	0.36
Neutral	40	0.14

Market Classification Logic:

- Bull Market: Return > +1%
- Bear Market: Return < –1%
- Neutral Market: –1% ≤ Return ≤ +1%

Each month's return was categorized based on these thresholds.



NIFTY 50 Closing Price (2000–2025)

X-axis → Date (2000–2025)
Y-axis → Closing Price
Shows trends, 2008 crash, 2020 dip, and recent growth.
Purpose: visually shows market uncertainty over time.

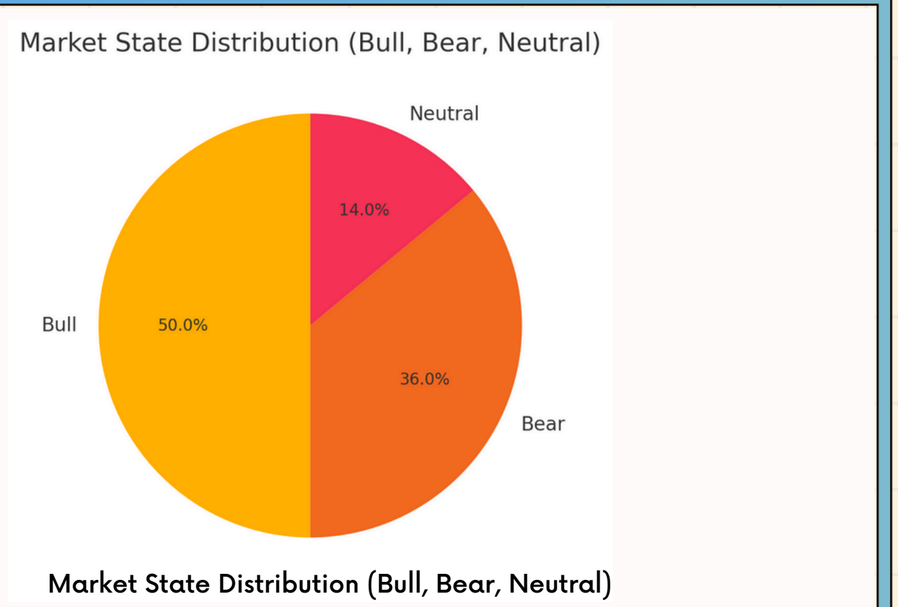
Market State Classification

Market Classification Logic:

- Bull Market: Return > +1%
- Bear Market: Return < −1%
- Neutral Market: −1% ≤ Return ≤ +1%

Each month's return was categorized based on these thresholds.

Category	Value
Bull	50%
Bear	36%
Neutral	14%



Market State Distribution (Bull, Bear, Neutral)

Interpretation:

- The NIFTY 50 spent half the time (50%) in bullish phases.
- Around 36% of months were bearish, showing notable volatility.
- Only 14% were neutral, meaning the market rarely stays stable.
- These probabilities are used in further slides for EMV and Bayesian analysis.

Dataset Overview

NIFTY 50 (2000–2020) highlighting 2008 crash & 2020 COVID dip.

Metric	Value
Mean Monthly Return	0.96%
Standard Deviation	6.70%
P(Bull Market)	0.5
P(Bear Market)	0.36
P(Neutral)	0.14

Dataset Used:

- Source: NIFTY 50 Index (2000–2025)
- Records: 5,061 daily observations
- Columns: Date, Open, High, Low, Close, Volume
- Derived metrics: Monthly Returns, Rolling Volatility

Payoff Matrix (3 Strategies × 3 Market States)

Strategy	Bull	Bear	Neutral
Buy & Hold	0.035	-0.025	0.005
Momentum	0.028	-0.010	0.007
Volatility-Based	0.020	-0.005	0.003

<u>Interpretation:</u>
Payoffs are average monthly returns for each strategy under each market condition.
Each number represents possible "reward" (profit/loss).

Solving – Decision Making Models

Maximin (Pessimistic Approach)

Choose strategy with best of the worst outcomes.
→ Worst payoffs:

- BH = -0.025
- MOM = -0.010
- VOL = -0.005

→ Highest (best) among worst = -0.005 → Volatility Strategy
Best for risk-averse investors.

Maximax (Optimistic Approach)

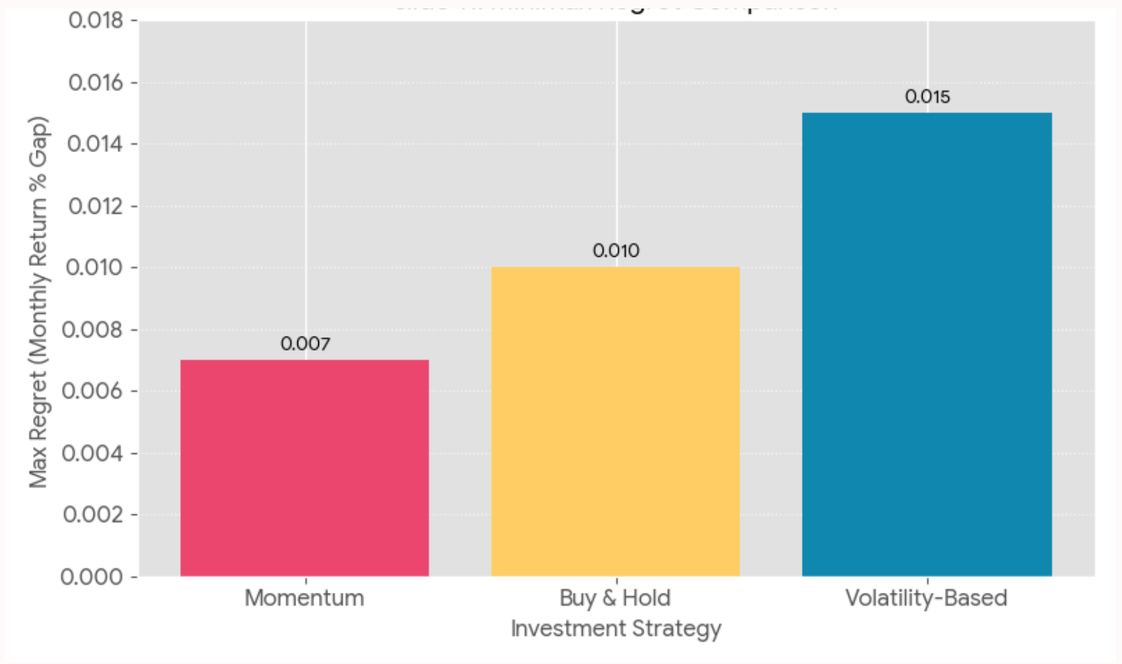
Choose strategy with best of the best outcomes.

→ Best payoffs:

- BH = 0.035
- MOM = 0.028
- VOL = 0.020
- → Highest = 0.035 → Buy & Hold

Best for risk-seekers.

Minimax Regret



Momentum Strategy has smallest max regret (0.007) → Best choice for balanced investors.

Regret = (Best payoff in that state – actual payoff)

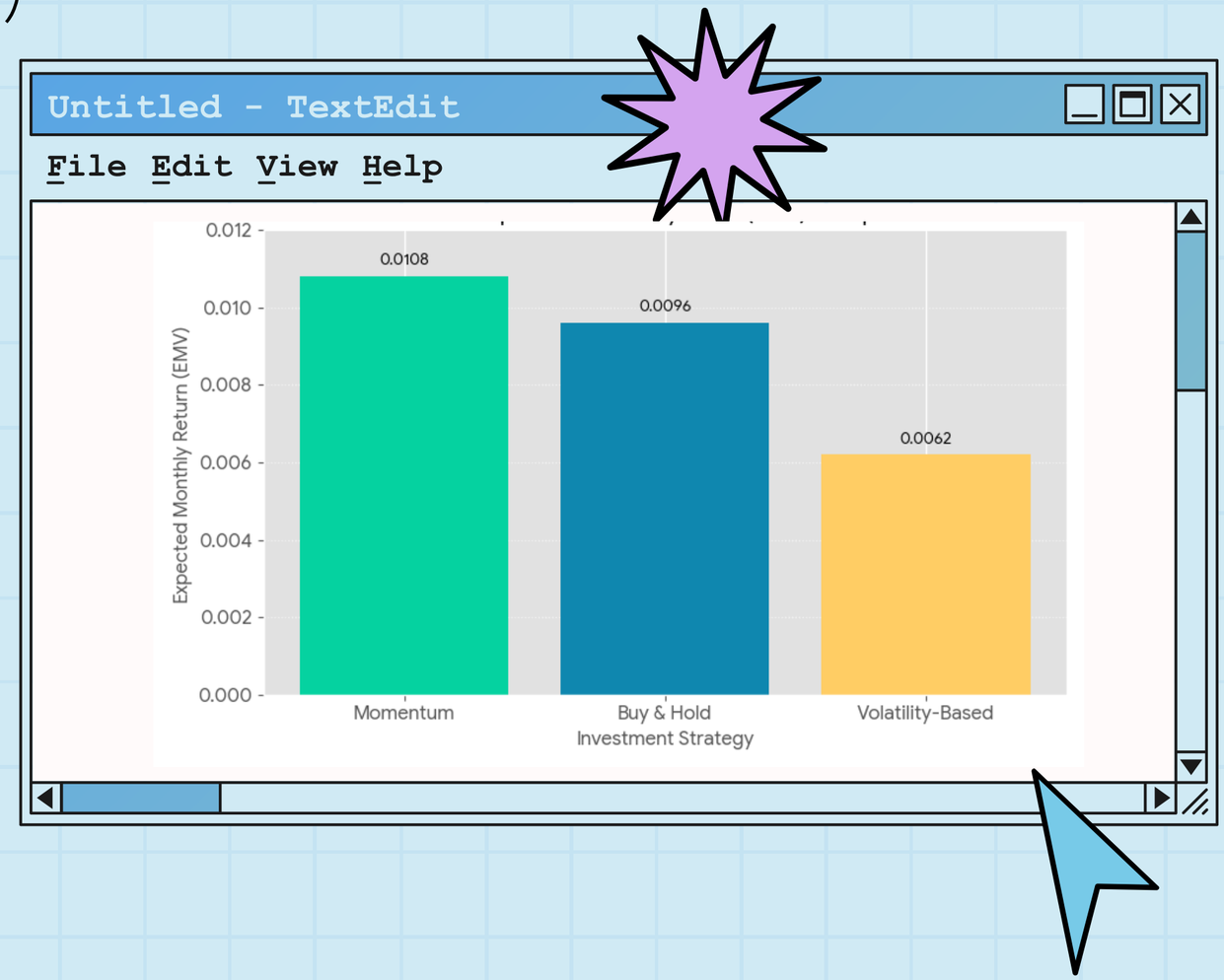
Strategy	Max Regret
BH	0.010
MOM	0.007
VOL	0.015

Expected Monetary Value (EMV)

$$EMV = \sum(Payoff \times Probability)$$

Strategy	EMV Calculation	EMV
Buy & Hold	(0.035×0.50) + (-0.025×0.36) + (0.005×0.14)	0.0096
Momentum	(0.028×0.50) + (-0.010×0.36) + (0.007×0.14)	0.0108
Volatility	(0.020×0.50) + (-0.005×0.36) + (0.003×0.14)	0.0062

Best Strategy = Momentum (Highest EMV = 0.0108)



Bayesian Updating

Suppose a new positive momentum signal (evidence) appears.

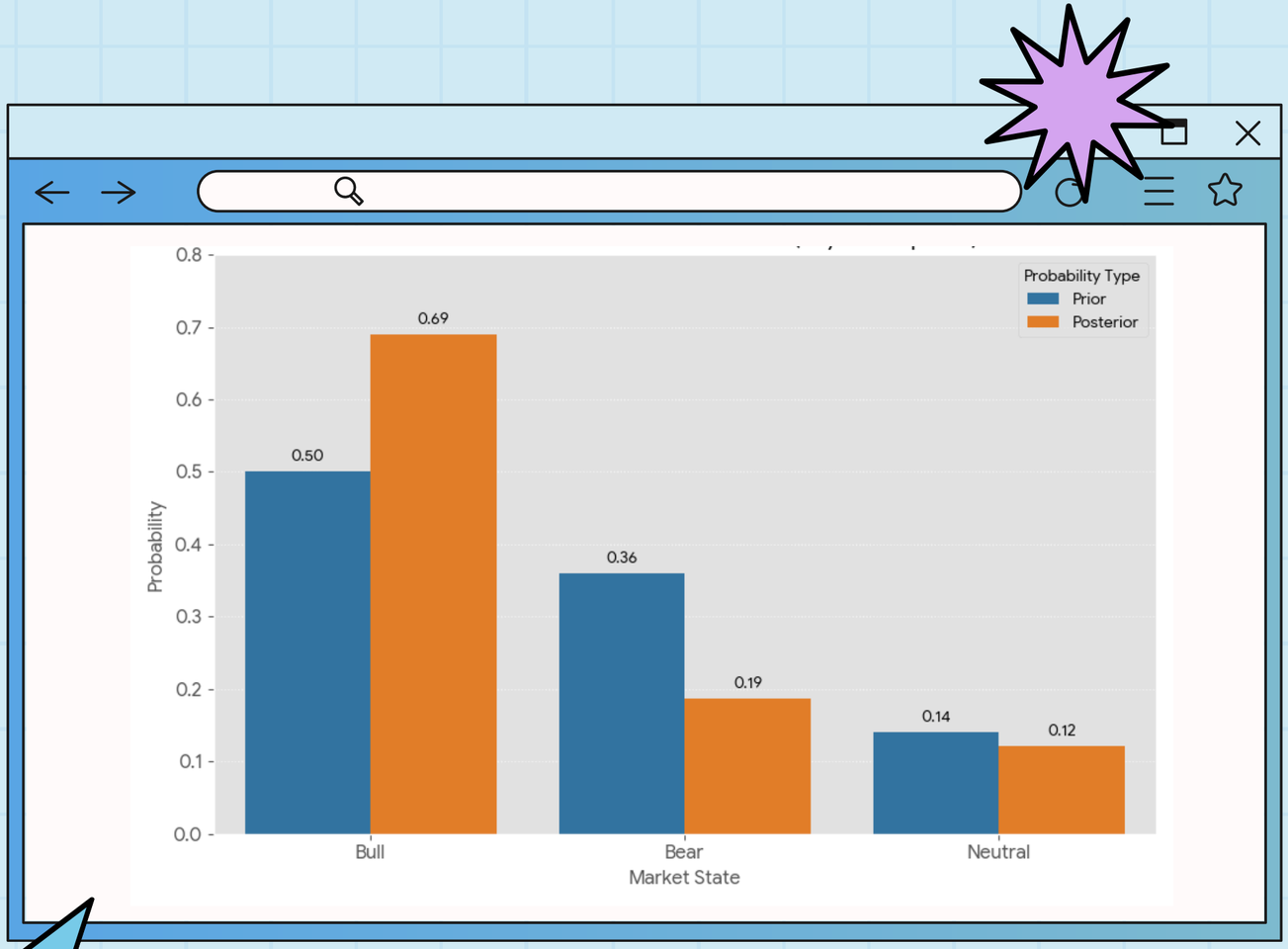
<u>Prior probabilities:</u>
P(Bull) = 0.50, P(Bear) = 0.36,
P(Neutral) = 0.14

<u>Conditional probabilities of signal:</u>
P(Signal|Bull) = 0.8
P(Signal|Bear) = 0.3
P(Signal|Neutral) = 0.5

$$P(Bull|Signal) = \frac{P(Signal|Bull) \times P(Bull)}{P(Signal)}$$

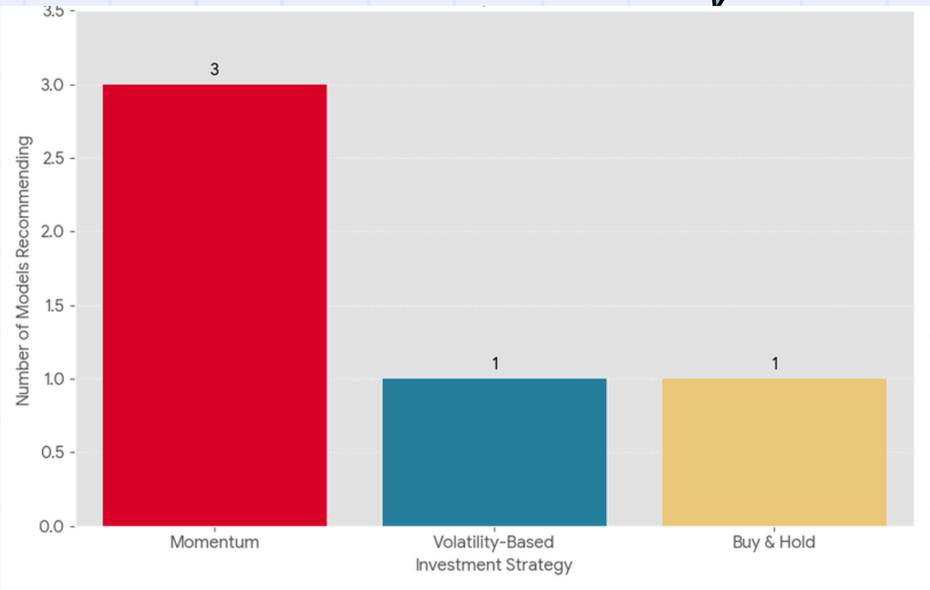
$$P(Signal) = (0.8 \times 0.50) + (0.3 \times 0.36) + (0.5 \times 0.14) = 0.58$$

$$P(Bull|Signal) = \frac{0.8 \times 0.50}{0.58} = 0.69$$



Posterior P(Bull) = 0.69 (up from 0.50)
→ Market is now more likely bullish → Momentum strategy becomes even stronger.

Integrated Analysis & Results



Reason:

Performs best on average (EMV)
Lowest regret
Strong after Bayesian update
Balances risk and reward effectively

Model	Best Strategy
Maximin	Volatility
Maximax	Buy & Hold
Minimax Regret	Momentum
EMV	Momentum
Bayesian	Momentum

Final Choice → Momentum Strategy

Real-World Application & Conclusion

Applications:

- Used by mutual fund managers and data-driven investors.
- Helps avoid emotional decision-making.
- Can be applied to business forecasting, pricing, and project investments.

Conclusion

Momentum strategy gives consistent and rational results across models.
Statistical decision-making reduces risk and improves long-term success.
In an uncertain world, data-backed models beat intuition.

"Good decisions come from data; great ones come from learning."

Thank you!

To our lovely Professor
Dr. Khinal Parmar